Hyunah Lim

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Education

Ph.D. in Mathematics, University of Maryland, 2019-2026.
Dissertation defended December 2025; degree expected Spring 2026.

M.S. in Mathematics, Yonsei University, 2017-2019.

B.A. Double Major in Economics and Applied Statistics, Minor in Mathemtaics, Yonsei University, 2012-2017.

Preprints and Publications

Modeling the Impact of Dynamic Hypoxia and Reoxygenation on Radiotherapy Response in Glioblastoma.
Hyunah Lim and Abba B. Gumel
In preparation

Age-structured mechanical models for tumor growth.
Doron Levy, Hyunah Lim, Antoine Mellet, and Maeve Wildes
Journal of Mathematical Biology **92** (2026), 63. DOI:10.1007/s00285-026-02389-z

Mathematical assessment of the roles of vaccination and Pap screening on the burden of HPV and related cancers.
Soyoung Park, Hyunah Lim, and Abba B. Gumel
Bulletin of Mathematical Biology **87**(12) (2025), 182. DOI:10.1007/s11538-025-01548-5

Theses

Age-structured models of tumor growth and evolution of drug resistance.
Ph.D. dissertation University of Maryland, Department of Mathematics, 2025

Some basic understandings of the Lascar Group: Lascar Group is a topological group.
Master's thesis Yonsei University, Department of Mathematics, 2019 RISS

Talks

Age-structured mechanical models for tumor growth (Lightening talk). Summer School on Algorithmic Cancer Biology, National Cancer Institute, July 2026.

Age-structured models of tumor growth and evolution of drug resistance. Doctoral Dissertation Defense, University of Maryland, Dec 2025.

Age-structured mechanical models for tumor growth. RIT on Mathematics of Infectious Diseases, University of Maryland, Nov 2025.

Mathematical modeling of the impacts of screening and vaccination in HPV (Poster talk). Mathematical Opportunities in Digital Twins, George Mason University, Dec 2023.

Mathematical modelling of the impacts of screening and vaccination in HPV (Poster talk). Predictive Modeling in Biology and Medicine Conference, University of California, Riverside, Nov 2023.

Introducing approaches to tumour growth models. Preliminary Oral Examination, University of Maryland, Mar 2023.

Lascar group is a topological group. Master's Thesis Defense Presenatation, Yonsei University, Dec 2018.

Awards

- Ruth-Davis Fellowship, University of Maryland, 2019-2024
- Dean's Fellowship, University of Maryland, 2019-2021.
- Academic Excellence Scholarship, Yonsei University, 2015

Teaching & Mentoring Experience

MathBio Summer Camp: Foundations of Mathematical Biology

Instructional Team Member (July 2026)
University of Maryland Institute for Health Computing (UM-IHC)

- Delivered lectures on tumor biology and mathematical oncology.
- Developed instructional materials on classic mathematical models of tumor growth.
- Designed and mentored a research project on tumor growth and resistance evolution under chemotherapy.

Brin Summer School: OneMath World School

Teaching Assistant, Tutorial Program: Oncology & Tumor Growth Modeling (June 2026)
University of Maryland

University of Maryland

- Instructor, Math 120 - Elementary Calculus I (Current, Summer 2025)
- Teaching Assistant, Math 240 - Linear Algebra (Fall 2021, Fall 2023, Fall 2024, Spring 2025, Fall 2025).
- Teaching Assistant, Math 120 - Elementary Calculus I (Fall 2019, Spring 2020, Spring 2022, Spring 2024).
- Teaching Assistant, Math 140 - Calculus I (Spring 2023)
- Teaching Assistant, Math 141 - Calculus II (Fall 2020, Fall 2022).
- Teaching Assistant, Math 115 - Precalculus (Spring 2021).

Yonsei University

- Grader, MAT 2104 - Set Theory (Spring 2019)
- Teaching Assistant, MAT3109 - Abstract Algebra I (Spring 2018)

- Teaching Assistant, MAT1012 - Mathematics for Engineers II (Spring 2018, Fall 2018)
- Teaching Assistant, MAT1002 - Calculus and Vector Analysis II (Fall 2017)
- Teaching Assistant, MAT1001 - Calculus and Vector Analysis I (Spring 2017, Fall 2018)
- Teaching Assistant, MAT1011 - Mathematics for Engineers I (Spring 2017, Fall 2017)

Languages

English (fluent); Korean (fluent); Japanese (Basic).